

Software Requirements Specifications

EduWizard

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1. Introduction

1.1 Purpose of Software Requirements Specifications

The purpose of this document is to define the software requirements and specifications of a computer science course site builder. It is intended to provide an overview of the software's main functionalities from the perspective of the primary use cases and the solutions developed to satisfy those use cases.

The intended audience of this document are developers and, to a lesser extent, customers of the product. Developers may consult this document as a reference for the software requirements and specifications that they need to implement. Customers may also reference this document as a technical overview of the requirements and specifications of the application that they are using.

The readers are assumed to have basic knowledge of software requirements and specification tools such as UMLs, use case diagrams, domain models and state machines.

1.2 Scope of Product

The name of the software application described in this document is EduWizard. The primary goal of EduWizard is to make the process of delivering online courses easier for CS Educators, and by extension the students they are teaching. To that end, EduWizard helps streamline the process for online course delivery because students have experienced many courses where content and assignment submissions are spread among many distinct platforms. EduWizard will create a platform where professors can build their own course sites that supports posting lectures and other course content. EduWizard will also allow professors to add assignment submission and grade posting to the site they create. By including each of these features, EduWizard can help professors create a central location for all their online course delivery needs.

EduWizard is limited in that it is only a course delivery platform. All content delivered by EduWizard must be first created by computer science educators. EduWizard is not an online school service and does not replace the need for computer science educators. Instead, EduWizard is a tool for computer science educators to use to streamline and enhance their online course delivery experience.

1.3 Definitions, Acronyms, Abbreviations, and Notation Conventions

- Definitions:
 - EduWizard: A computer science course site builder
 - Application: The application (sometimes used interchangeably with EduWizard)
 - Software Requirements Specification: This is a document of a software system, including the descriptions of functional and non-functional requirements of the system.
 - User Interface: This is the space that allows users to interact with machines.
 - Users: End users of the application. This group can be divided into two main subgroups who are the target audience of the application:
 - CS Students: In the scope of this document, “CS Students” are defined as university-level undergraduate students of the University of Waterloo enrolled in Computer Science courses. Sometimes also referred to as “student” or “students”.
 - CS Educators: In the scope of this document, “CS Educators” are defined as professors or teaching assistants of the University of Waterloo teaching Computer Science courses. Sometimes also referred to as “educator”, “professor”, or “teaching assistant”.
 - Login: This refers to logging into the EduWizard application. A user logs in using an email and password.
 - Logout: This refers to the action of a user knowingly exiting the application.
 - WCAG 2.0: The range of recommendations for making Web content more accessible. These guidelines help make content accessible to a wider range of people with disabilities, including blindness and low vision, deafness and hearing loss, learning disabilities, cognitive limitations, limited movement, speech disabilities, photosensitivity and combinations of these.
- Acronyms:
 - AODA: Accessibility for Ontarians with Disabilities Act
 - CS: Computer Science
 - SRS: Software Requirements Specification
 - TA: Teaching Assistant
 - UI: User Interface
 - UML: Unified Modelling Language
 - WCAG: Web Content Accessibility Guidelines

2. Overall Description

This part of the document includes the overall description of the application, including all factors that affect the application's requirements, the top three use cases of the application, and the characteristics of users who intend to use EduWizard.

2.1 Product Perspective

The application works as a course site builder for CS Educators to provide a highly-customizable way to build course sites for their students. The most important function of EduWizard is for CS Educators to manage their course sites to suit all their needs without the need for additional platforms. Other additional features of EduWizard include student interaction with the built course sites, the ability to dispute grades in one place.

The following is a use case diagram that illustrates how EduWizard interacts with its actors, and its three main use cases.

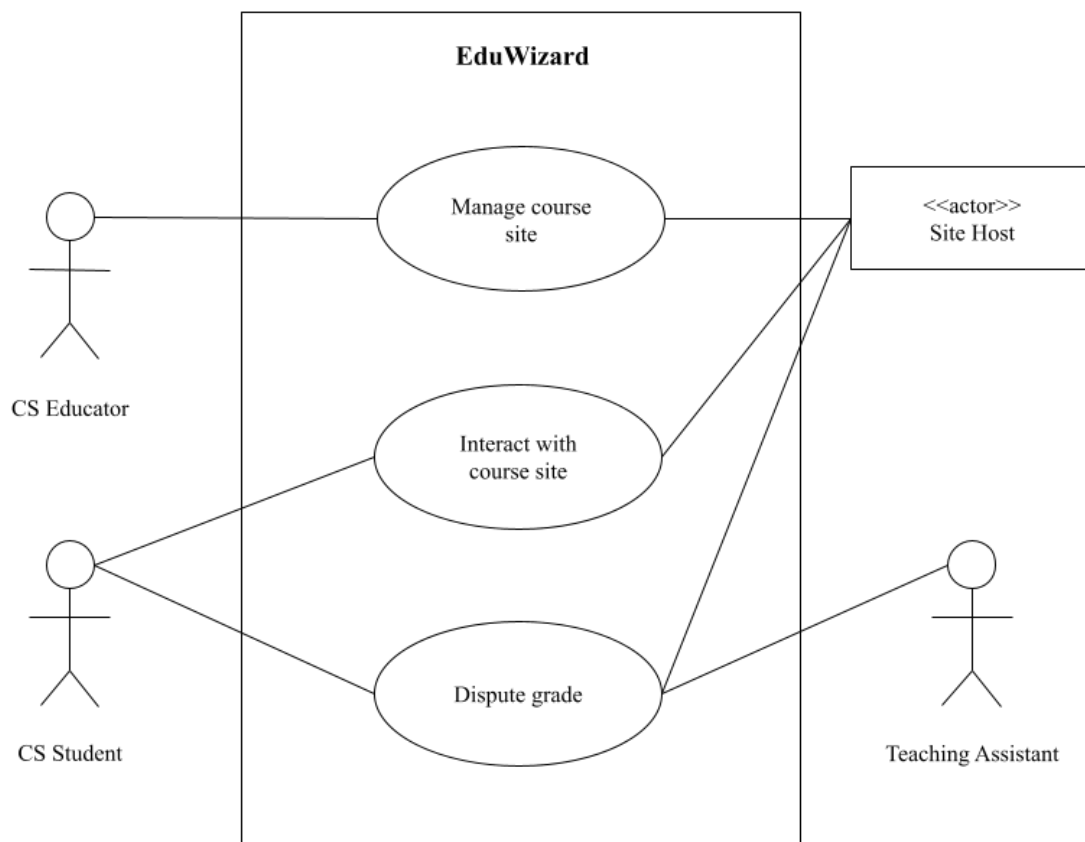


Figure 1: Use Case Diagram

Actor Descriptions

- **CS Educator**
This is the primary human actor of EduWizard. CS Educators use our application to build their course sites, this includes creating lectures, assignments, quizzes, and announcements, and reviewing and posting grades.
- **CS Student**
This is the secondary human actor of EduWizard. CS Students use our application to view built course sites they have an account for. This includes viewing lectures, assignments, announcements, and grades, and completing quizzes.
- **Site Host**
The site host enables the instructor to put their website up for users to access. This is integral for all instructors because they will have control of what they put up.
- **Teaching Assistant**
This is also a secondary human actor of EduWizard. Teaching Assistants use our application to edit course sites, but with limited functionality. This includes grading assignments and interacting with grade disputes.

Use Case Descriptions

- **Manage course site**
CS Educators need to have control over what content can be displayed to students and TAs etc. They have control over due dates, office hours, and other important elements of the course.
- **Interact with course site**
Students need to be able to interact with the course site to view content and complete assignments. Students should be able to complete quizzes, submit assignments, and view lectures by interacting with the course site.
- **Dispute grade**
Students who find that there is something wrong with their grade can easily dispute it and TA's can easily view the dispute. A student can submit a form with the questions they think are incorrectly graded, along with a description of why they think they're right. The TA will then receive this and can either accept or decline the reasoning, add their own reasoning, and update the grade on the spot.

2.2 Product Functions

This part of the document covers the workflow diagrams and user stories for the top three use cases and the wants and needs of each use case derived from user stories.

By conducting several interviews and stakeholder analyses of our most important user, the top three use cases of our application were discovered to be:

1. Manage Course Site
2. Interact with Course Site
3. Dispute Grade

The diagrams and corresponding descriptions in this section and the following sections will solely focus on these three key use cases.

2.2.1 Workflow Models

The workflow model is a graphical representation of a series of tasks that are performed to accomplish one or more of our use cases. The UML activity diagram is a great way to demonstrate the workflow.

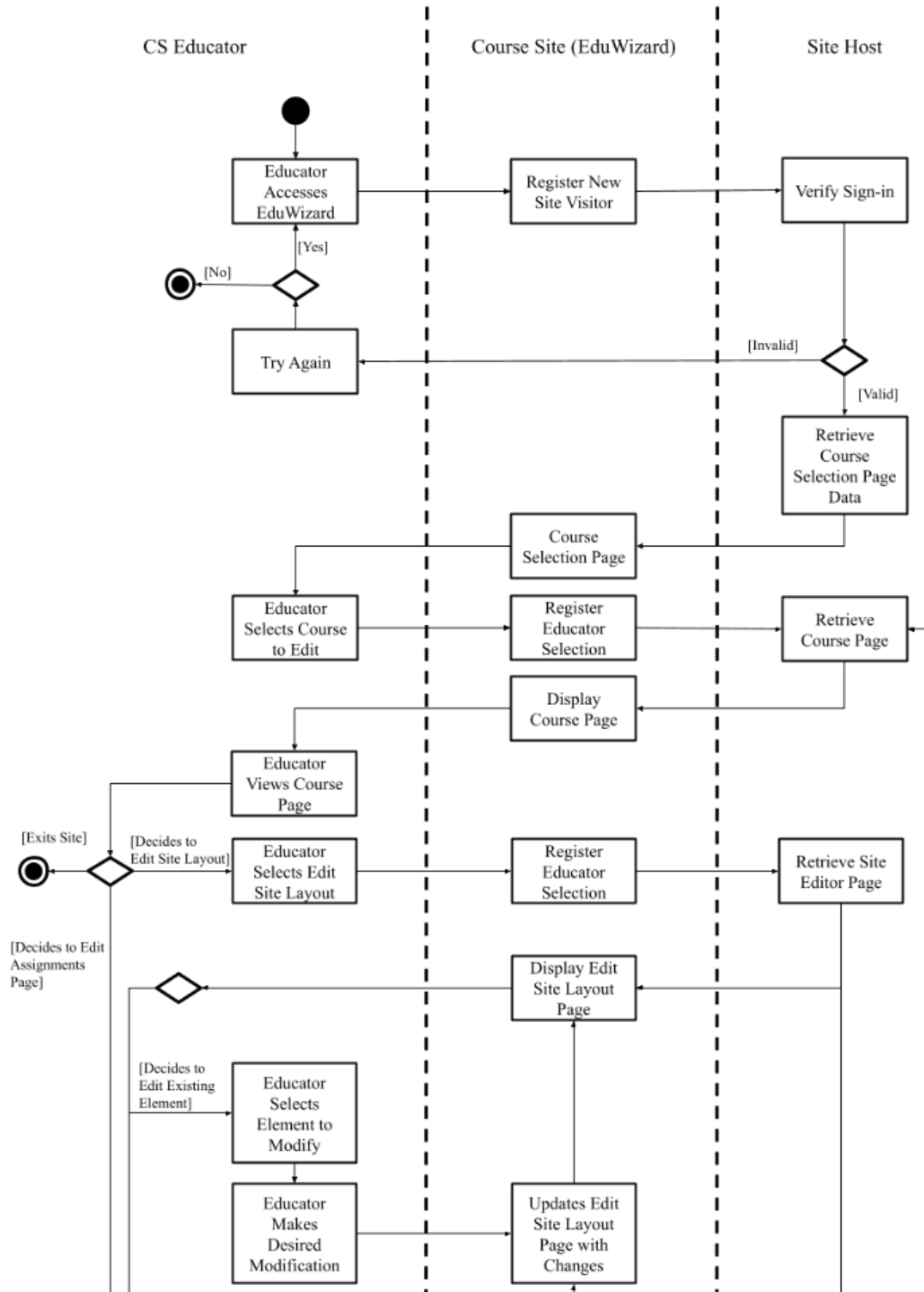
The following UML activity diagrams demonstrate the workflow of the top three use cases of our product:

Manage Course Site

This diagram models one of the key features of our platform, which is the manage course site flow. The flow has 3 actors: the CS Educator, the Course Site (EduWizard), and Site Host. We have grouped some of the potential scenarios in the use case to simplify the workflow diagram. There are many things that a CS Educator can do when managing their site including, adding/editing assignments, lectures, announcements, and quizzes, viewing grades and grade disputes, editing the site layout, and adding new pages. To simplify the workflow diagram we only show the flow to edit the site layout and edit the Assignments page. We did this because editing the site layout involves a lot of interaction between our actors so we wanted to demonstrate what needs to happen behind the scenes to allow CS Educators to alter their site layout. We also grouped the adding/editing assignments, lectures, announcements, and quizzes cases together because each of these scenarios involves a very similar set of actions, so by modelling one scenario we can demonstrate the other scenarios with fairly high accuracy.

When a user wants to manage their site, the first thing they are going to do is open (sign into) EduWizard, where they can select a course site to edit. The home page could contain announcements, deadlines, etc, but our focus is on the grades. So from the home page, there will be an option to “Edit Site Layout” via an options menu. This will bring the educator to the site layout editor page where they can add, remove, and edit elements on their site.

The educator can also select the “Assignments” page requiring the course site to retrieve the assignment data associated with the course. The site will display all the created assignments for the educator to view. From here the educator can edit specific assignments by selecting them, or create a whole new assignment by clicking an add new assignment button. After they are done editing they can click the save button that will update their course site. They can then continue to edit other aspects of their course site or leave the site entirely.



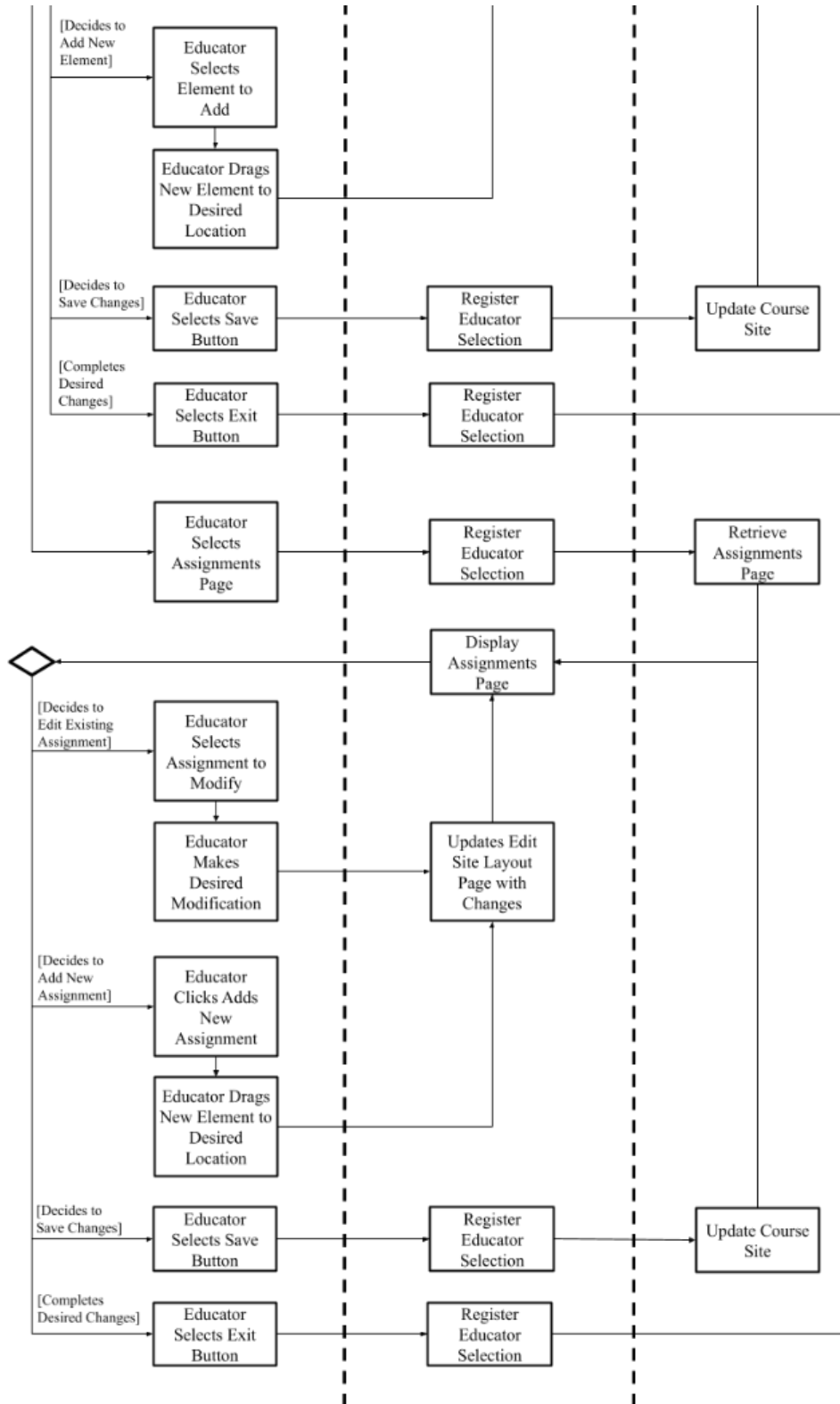


Figure 2: Manage Course Site Workflow Diagram

Interact with Course Site

The following diagram models the flow of a student interacting with our course site. The diagram models the key use cases - viewing lectures, accessing assignments and submitting assignments. Since there are various different use cases for a student interacting with the course site which would take a lot more space to model, we decided to go with the most important.

When a student wants to interact with the course site, the first thing they are going to do is open (sign into) the course site, which requires the course site to load home page data from the site host. Through the home page, they can navigate to the lectures and assignments page via a navigation menu.

If the student decides to view lectures they make their selection, then the course site retrieves the page from the site host and displays it to the student. Now, the student can select a particular lecture video to view. The course site will retrieve lecture video data from the site host and display it in a video player to the student.

Similarly, when a student navigates to the assignments page, they can decide to either view or submit an assignment. For viewing, the course site will retrieve the assignment information page data from the site host and display it to the student. For submissions, the course site will display the assignment submission page to the student. Here the student can create a submission and submit. The course site will register the submission and send it to the site host. The site host will then store the submission.

At any stage of interacting with the course site, the student can choose to exit the site which will end the workflow.



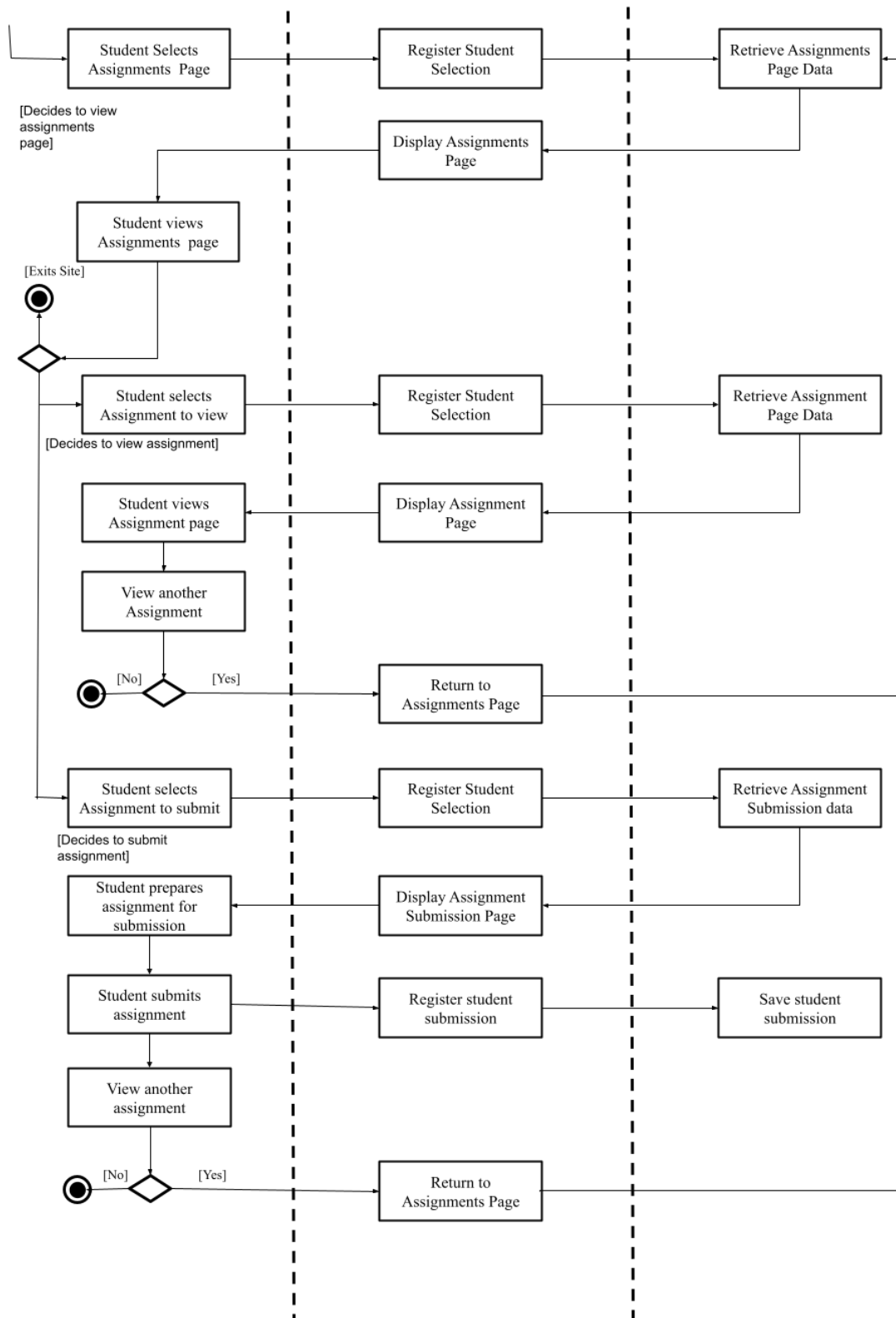


Figure 3: Interact with Course Site Workflow Diagram

Dispute Grade

This diagram has three actors: The student, the TA and EduWizard. When a student wants to start a grade dispute they can do so from their assignment submission page. The student will fill out a form with which questions they want to dispute and the reasoning for that dispute. They will confirm that dispute or cancel. On cancel, they can go back to editing the dispute. On confirmation, EduWizard will update the submission to show the grade dispute, and notify the TA. From here the TA will be able to review, accept, and decline the dispute, and will be able to submit the results of the dispute. On cancel, they can go back to editing and on confirmation, EduWizard updates the submission with the results of the dispute and notifies the student. From here the student can either accept or decline the results, allowing them to dispute again. Our diagram has a limitation where it doesn't show that a student can only dispute a certain number of times, as a function of time or can not redispute a mark. These are all possible use cases, we chose the most general to give the scope of the use case itself. We chose an activity diagram as data is contained to EduWizard in this use case, meaning that a data flow diagram would not show the full scope of this use case.

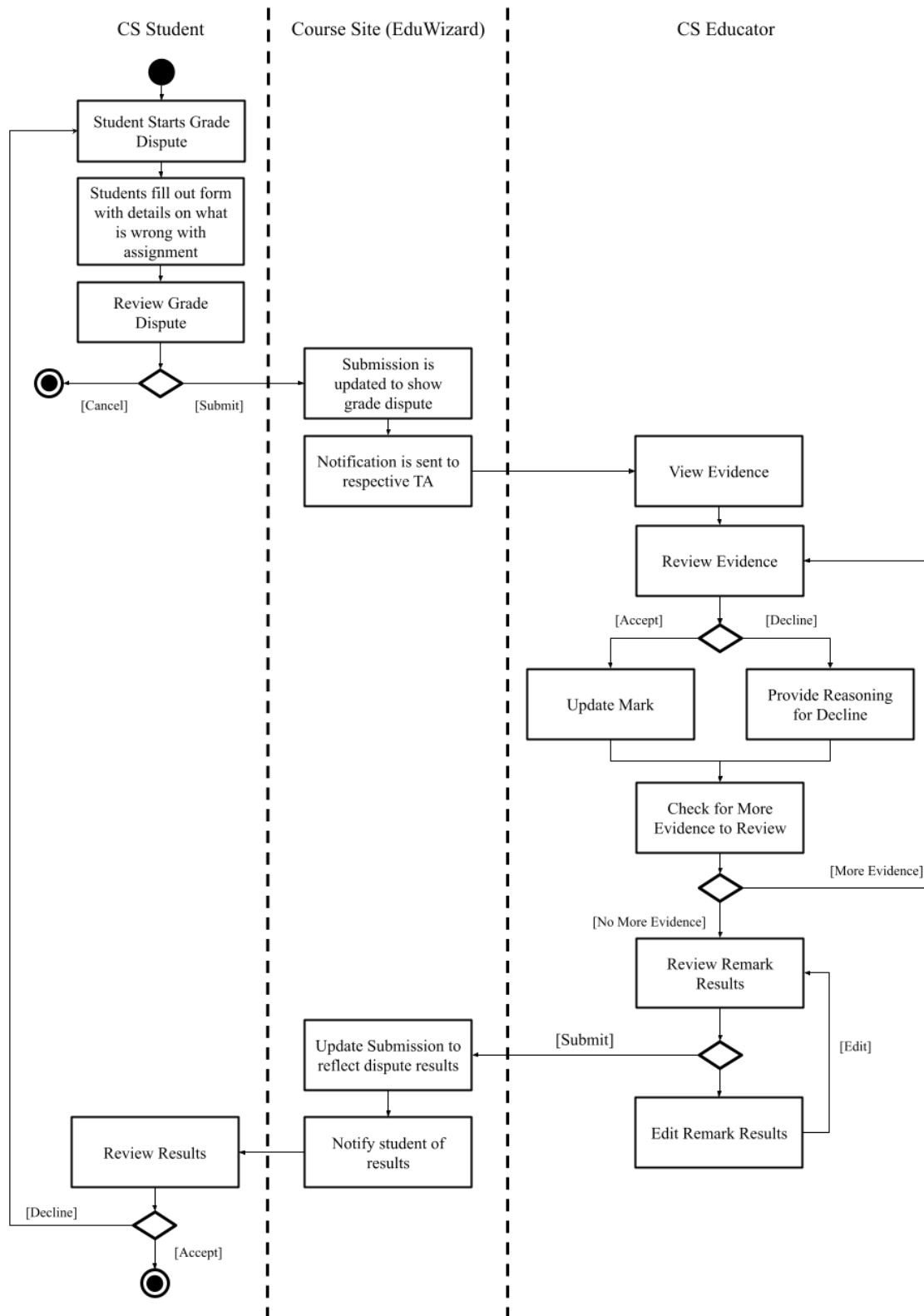


Figure 4: Dispute Grade Workflow Diagram

2.2.2 User Stories

User stories are a lightweight method used to determine the requirements of use cases from the perspective of users, they help identify useful features and the benefits that they bring. It is a short statement that is written from the point of view of the users about any new functionality or feature.

User stories have the following structured syntax:

As a **user/role**, I want **something**, so that **benefit** is achieved.

The following are the user stories for the top three use cases:

Managing Course Site

- As a CS Educator, I want an easy-to-use and secure online platform to upload my lectures, so that I can deliver my course content asynchronously to students.
- As a CS Educator, I want to have control over the look and feel of my course website, so that students find it easier to navigate through the website.
- As a CS Educator, I want to make important announcements related to the course content, so that my students can be notified of important information and upcoming deadlines.
- As a CS Educator, I want a visual way to build my course website, so that I don't have to deal with complicated bootstrap syntax.

Dispute Grade

- As a CS Educator, I want to be able to view all remark requests for a particular assignment in a centralized location, so I can view them all quickly with minimal search effort
- As a CS Educator, I want to be able to view the remark request with questions and evidence, so that I can get a clear picture of what the student feels the issue is with the assignment
- As a CS Student, I want to be able to select specific parts of the assignment I want to be re-marked, so that I can get tailored solutions and comments on that question.
- As a CS Student, I want to be able to create a remark request on the course site, so that I don't have to navigate to another platform to handle these requests

Interact with Course Site

- As a CS Student, I want to be able to easily view lectures, course notes and other course content on the website in an organized fashion so that I can see all of my course material in a single amalgamated place.
- As a CS Student, I want to be able to submit assignments with any file format on the website so that I can submit all my assignments in one place.
- As a CS Student, I want to be able to do quizzes on the website so that I can complete my assessments online.

2.3 User Characteristics

Users of EduWizard are required to have a valid email to use when creating or signing into their account.

The two main users of EduWizard are:

CS Educators:

The user of the EduWizard site builder platform is a CS Educator who is expected to be familiar with programming the front-end and back-end interface of a website and are familiar with hosting it on their own servers. They should be familiar with using browsers for accessing the built course site. CS Educators will have an email account that is linked to the login. They are also expected to have completed our software usage tutorial, which is a set of popups that familiarize them with the steps for the different use cases related to building and managing a course site.

CS Students:

The user of the built course site is a CS Student who is expected to be familiar with using a mobile or desktop browser and can access the built course site. CS Students are expected to have email accounts that they can use to create a login.

2.4 Constraints

- **Regulatory policies**
 - Personal information
 - Accessibility: WCAG 2.0 is a set of recommendations to follow, to help the site to be more accessible to people with disabilities, such as low vision and blindness.
- **Software/Infrastructure**
 - OS requirement for site host: must be a Linux, Windows or MacOS server.
- **Audit functions**
 - Checking who accessed the system for data integrity
- **Control functions**
 - Students should have the ability to control what personal information is visible for their profile
- **Criticality of the applications**
 - Data Leakage is a critical concern for the application.
- **Safety and security considerations**
 - The intellectual property of CS educators should not be leaked via our application or captured by our system, our security layer will only allow authenticated users to access the content.
 - Personally Identifiable Information of users must be encrypted and not shared with third parties or leaked.
 - Metrics such as grades will not be stored by our software and will remain private to authorized educators only.
- **Standards**
 - Our software must comply with Accessibility standards.
 - WCAG 2.0 is to be followed if we are in the public sector or are private and have 50+ employees, according to AODA.
- **Laws**
 - AODA (Accessibility for Ontarians with Disabilities Act)

2.5 Assumptions

When building a system, it is nearly impossible that by itself will satisfy all the requirements. Therefore, assumptions about how the environment behaves can be used.

The following are the assumptions of the top three use cases for EduWizard:

- **General Assumptions**
 - User is logged in with their account ID and password
 - User has the correct permissions to do the task they are attempting
 - A compatible browser is being used by user
- **Assumptions for Managing a Course Site**
 - General Assumptions
 - CS Educator's site host server is able to be accessed remotely by computers.
 - CS Educators have a functioning server to host their built course site on.
- **Assumptions for Disputing Grades**
 - General Assumptions
 - CS Educator updates student grades after marking.
 - CS Educator monitors disputes on their grade dispute page.
- **Assumptions for Interacting With a Course Site**
 - General Assumptions
 - CS Educators have added course material for students to interact with.

3. Specific Requirements

This section describes the requirements and specifications for EduWizard in detail. It contains the interface specification including domain models and UI sketches, scenarios, state machine models, and quality requirements.

3.1 Domain Model and User Interface Sketches

This section contains the detailed specification domain model and UI sketches of the three top-priority use cases for EduWizard, which are managing a course site, interacting with a course site, and disputing grades.

The domain model is shown in Figure 5, including the human actors (i.e., CS Educators and CS Students), non-human actors (i.e., Site Host), interfaces and other elements of the system and the associations between them,

The UI sketches are shown in Figures 6 to 10, and focus on the Manage Course Site and Dispute Grades use cases. Each sketch contains the UI elements (e.g., buttons, text boxes, and widgets) for each screen, a sample layout, and events that can be used as transitions in the navigation diagram shown in Figures 16 to 18.

3.1.1 Specification Domain Model

Below is the specification domain model diagram for EduWizard:

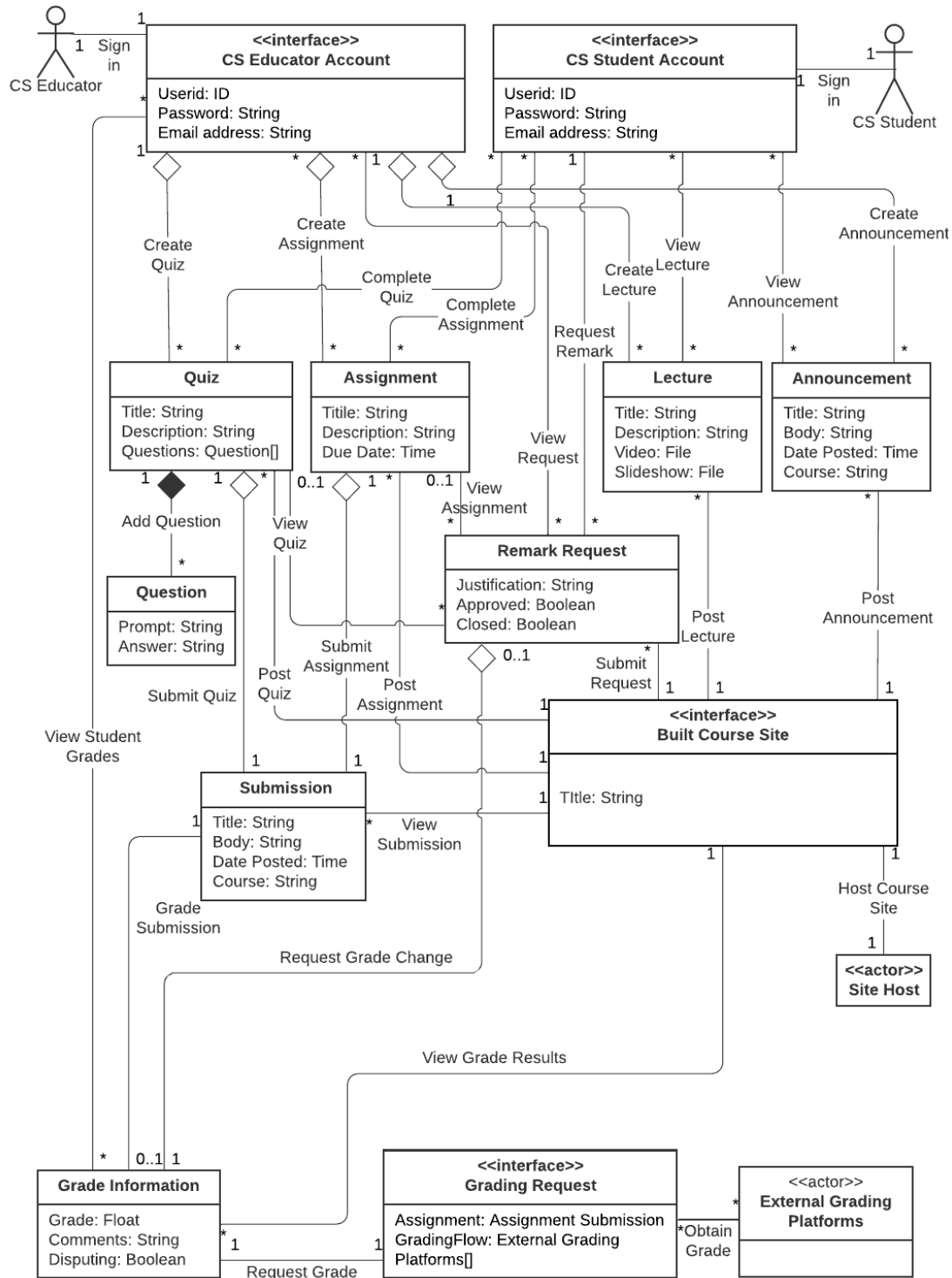


Figure 5: Specification Domain Model Diagram

Below are the descriptions of previous specification domain model diagram, including entity descriptions and relation descriptions

- **Entity Descriptions:**

- **Assignment:** An Assignment is similar to a Quiz, it has a Title and Description and a due date for the submission time.
- **Announcement:** Announcements have a title, body, date posted and for which course. It works as a notification to students that is displayed on the Built Course Site.
- **Built Course Site:** Used by CS Educators to post announcements, lectures, quizzes, assignments and grades. Contains the UI for CS students to access announcements, watch lectures, complete quizzes and submit assignments.
- **CS Educator Account:** An entity that stores CS Educator related information. It is used by CS Educators to log into the application with a userid and password. When first registering, CS Educators provide a 3rd party email account that is associated with the entity.
- **CS Student Account:** An entity that stores CS Student related information. It is used by CS students to log into the application with a userid and password. When first registering, CS students provide a 3rd party email account that is associated with the entity.
- **External Grading Platforms:** Includes adjacent systems that handle automated grading of CS assignments like Marmoset, Markus, etc.
- **Grade Information:** Used for Submissions, it contains a Grade, Comments for the Grade and a Boolean Dispute value to indicate if the grade is currently being disputed.
- **Grading Request:** Interface used by the application to send submissions to external grading platforms and receive the computed assignment grade for each submission.
- **Lecture:** A Lecture is the data required to show the student course content, in a video format, and also in Powerpoint/Slide format. A lecture holds the fields: Title, Description, Video, Slideshow.
- **Question:** Used in Quizzes, which contains both a Prompt and an Answer.
- **Quiz:** A Quiz consists of questions and has a title and description.
- **Remark Request:** Used by students to request a change to their Grade information.

- **Submission:** Stores information related to a CS student's assignment or quiz submission. Including the title, student response body, date posted and related course. A submission is mapped to an assignment or quiz posted by the CS Educator.
- **Site Host:** The built course site hosting server is owned and managed by the CS Educator.

● Relation Descriptions

- **Site Host:** The Site Host hosts the built course site so that it is accessible to the web
- **CS Educator creates and posts an announcement:** A CS Educator can post an announcement to the built course site via their account. A built course site can have many announcements.
- **CS Educator creates and posts a lecture:** A CS Educator can post a lecture to the built course site via their account. A course site can have many lectures.
- **CS Educator creates and posts an assignment:** A CS Educator can post an assignment to the built course site via their account. A course site can have many assignments.
- **CS Educator creates and posts a quiz:** A CS Educator can post an assignment to the built course site via their account. A course site can have many quizzes.
- **CS Educator reviews remark request:** A CS Educator can review a remark request via their account.
- **CS Student views lectures:** A CS Student views the lectures via their account.
- **CS Student views course site:** A CS Student views the announcements via their account.
- **CS Student submits assignment:** A CS Student submits an assignment via their account to external grading platforms.
- **CS Student completes quiz:** A CS student completes a quiz via the course site by answering each of the questions associated with the quiz
- **CS Student request remark:** A CS Student disputes their grades to the built course site via their account which will generate a remark request
- **CS Educator reviews remark request:** A CS Educator can review remark requests via their account
- **CS Educator modifies grades:** A CS Educator can modify CS Student grades via their account.

3.1.2 User Interfaces Sketches

The following are UI sketches for Eduwizard. For the sake of completeness, these sketches focus solely on the CS Educator side of our application. Our product is primarily geared towards CS Educators and they are the stakeholders we interviewed during our Solution Fit Interviews. We wanted to ensure that our solution fits the needs of the CS Educators as they will need to build the sites students will use, so if our site builder does not meet the needs of the CS Educators there will be no course site for the CS Students to use.

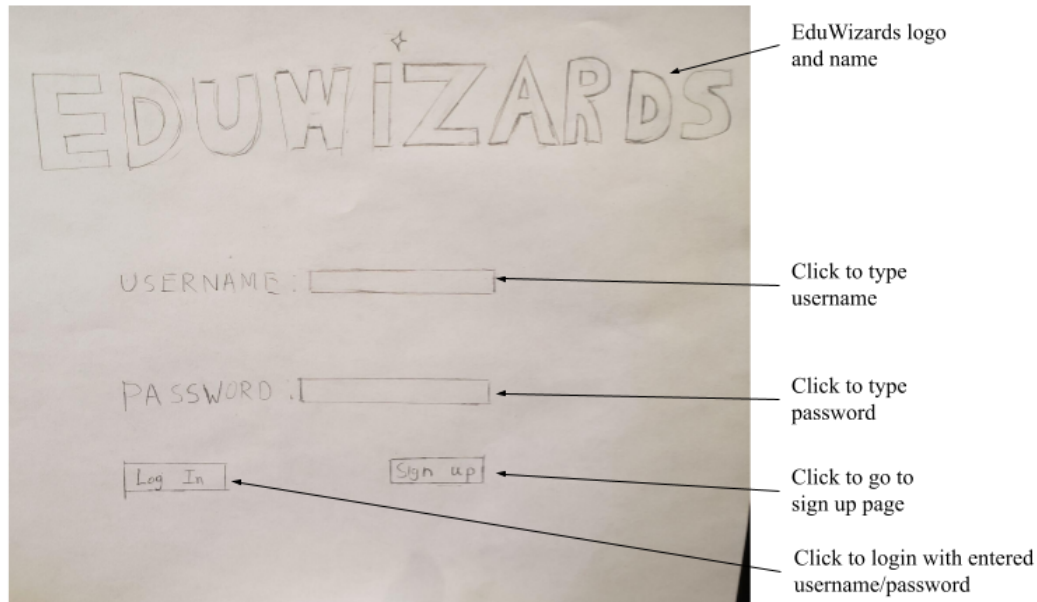


Figure 6: UI Sketch 1 - Login

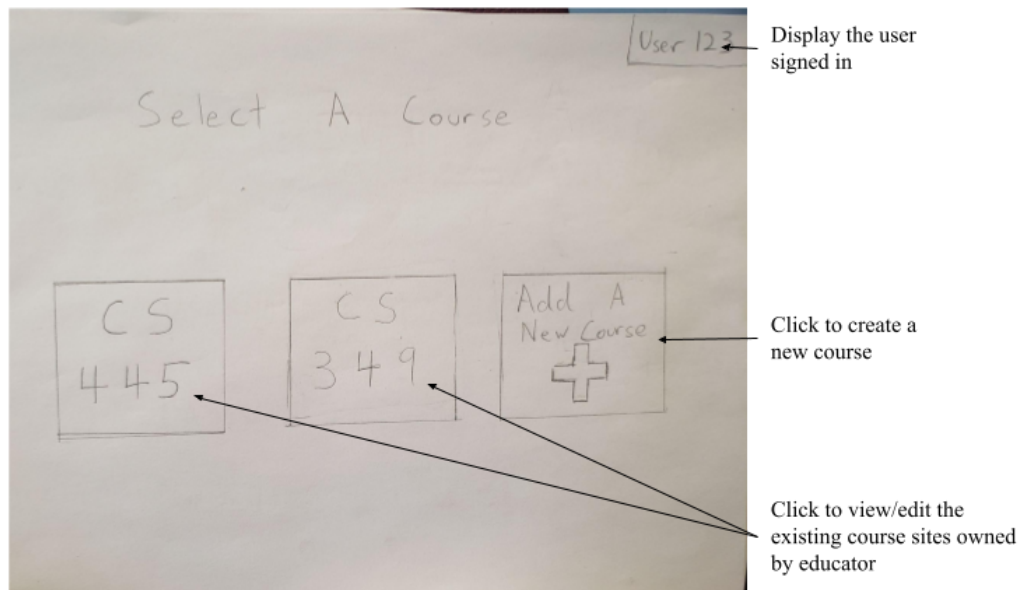


Figure 7: UI Sketch 2, Landing Page - Choose Course to Edit

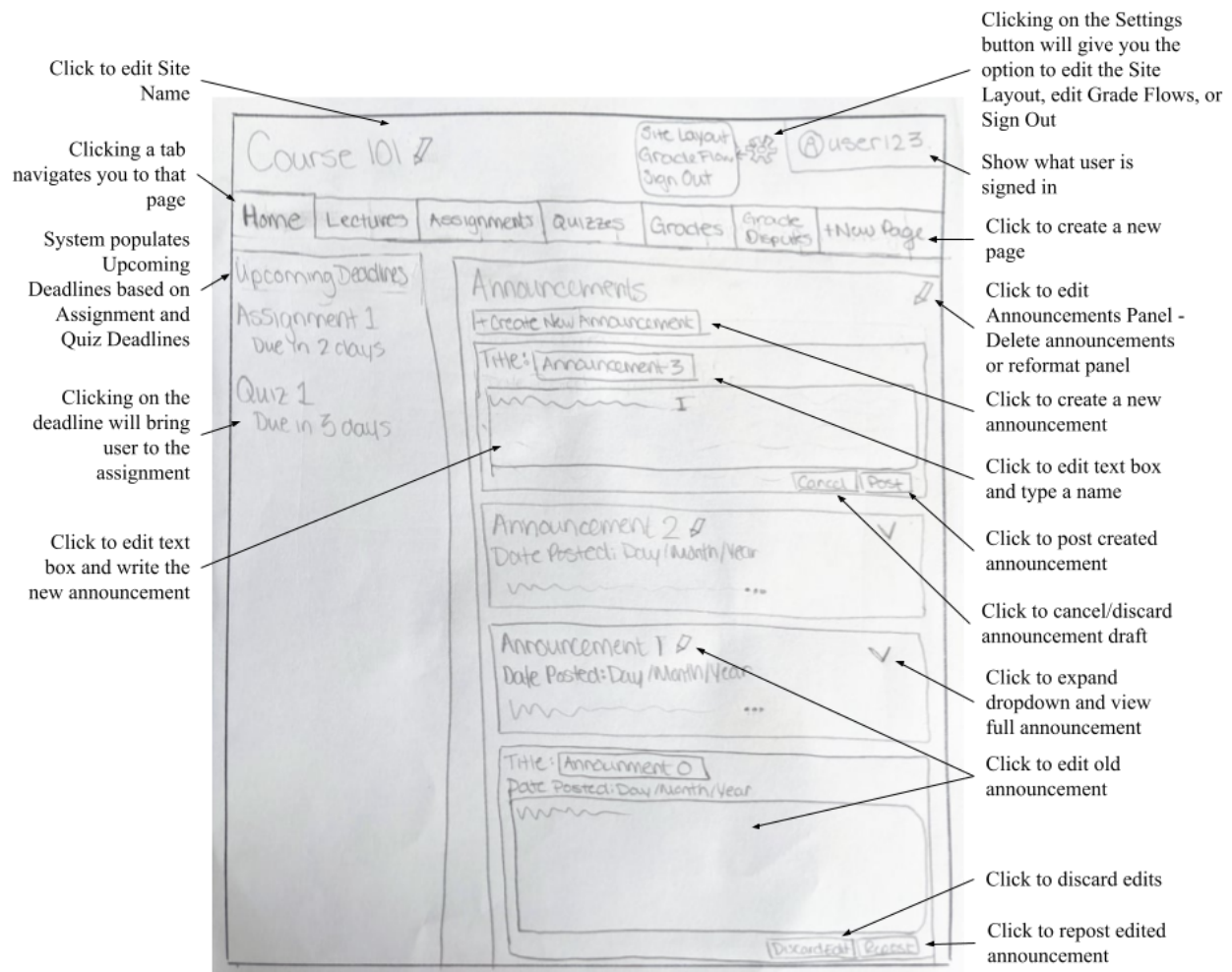


Figure 8: UI Sketch 3 - Course Home Page (Educator Side)

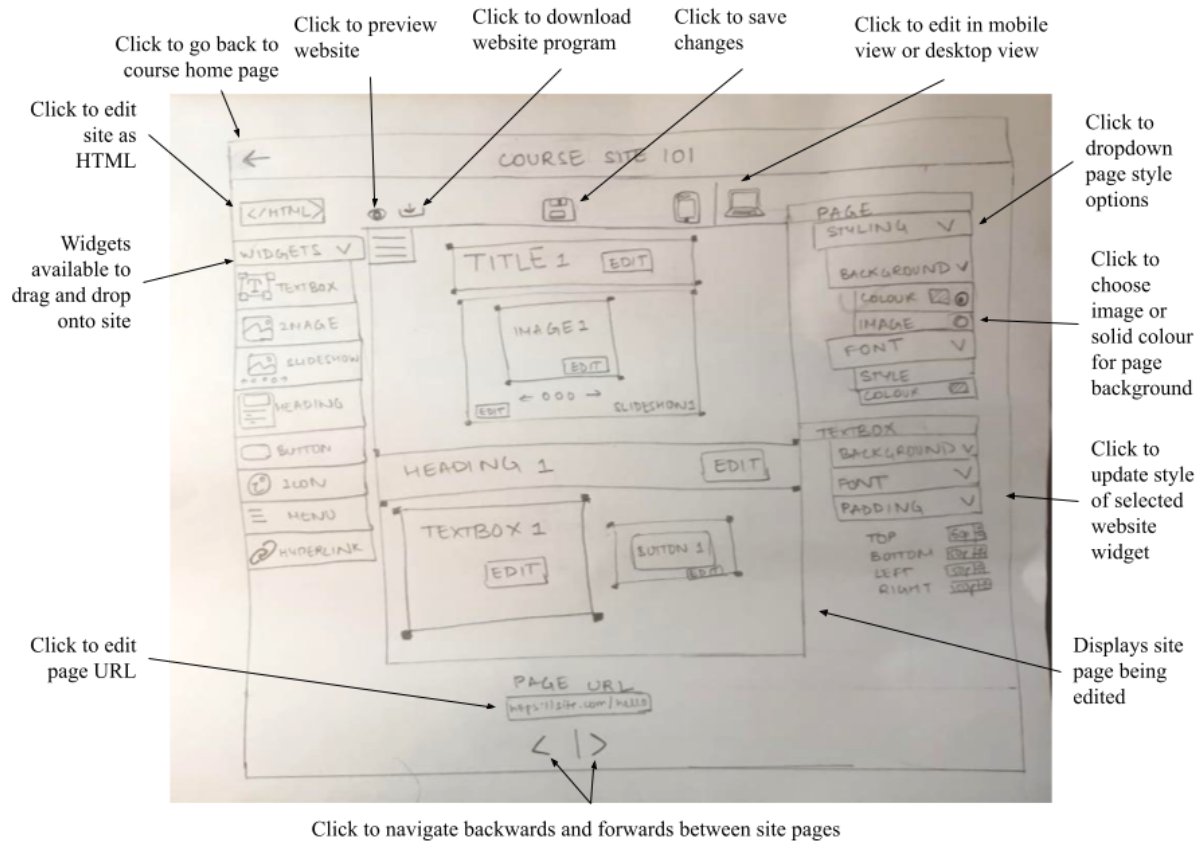


Figure 9: UI Sketch 4 - Site Editor

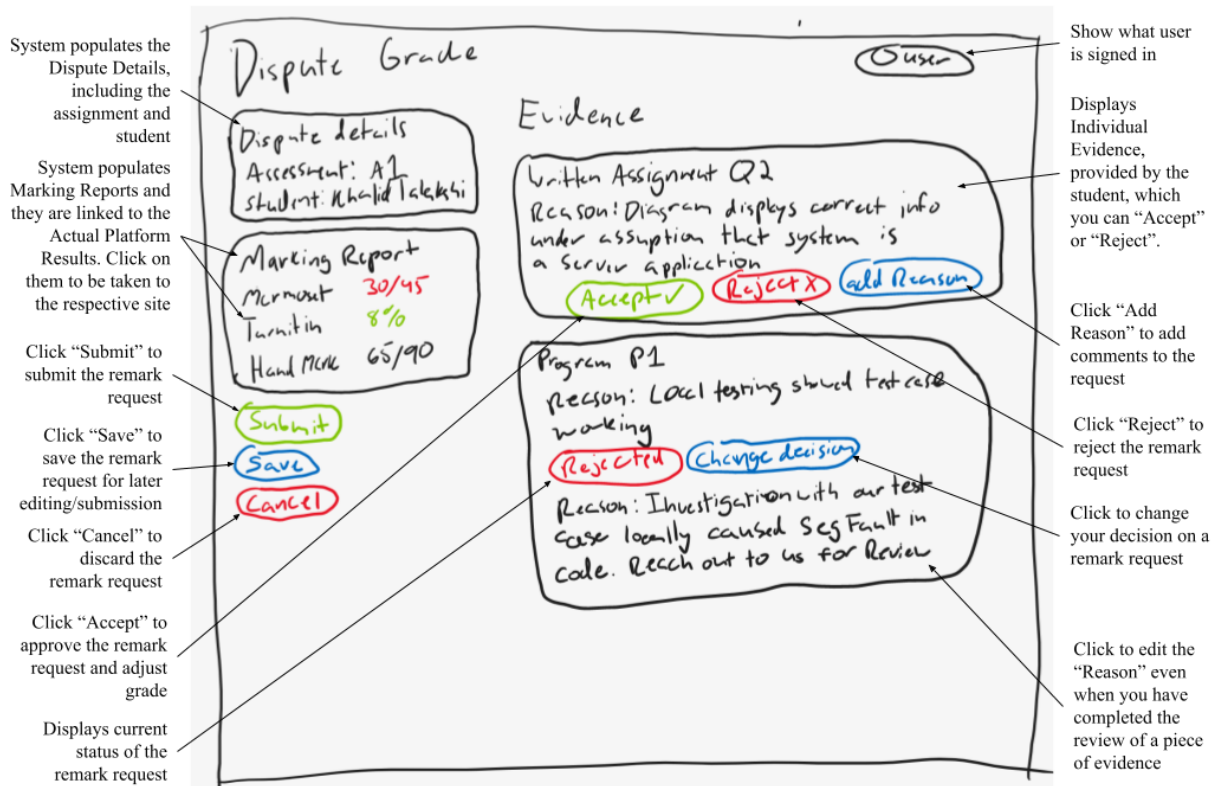


Figure 10: UI Sketch 5 - Dispute Grade Page (Educator Side)

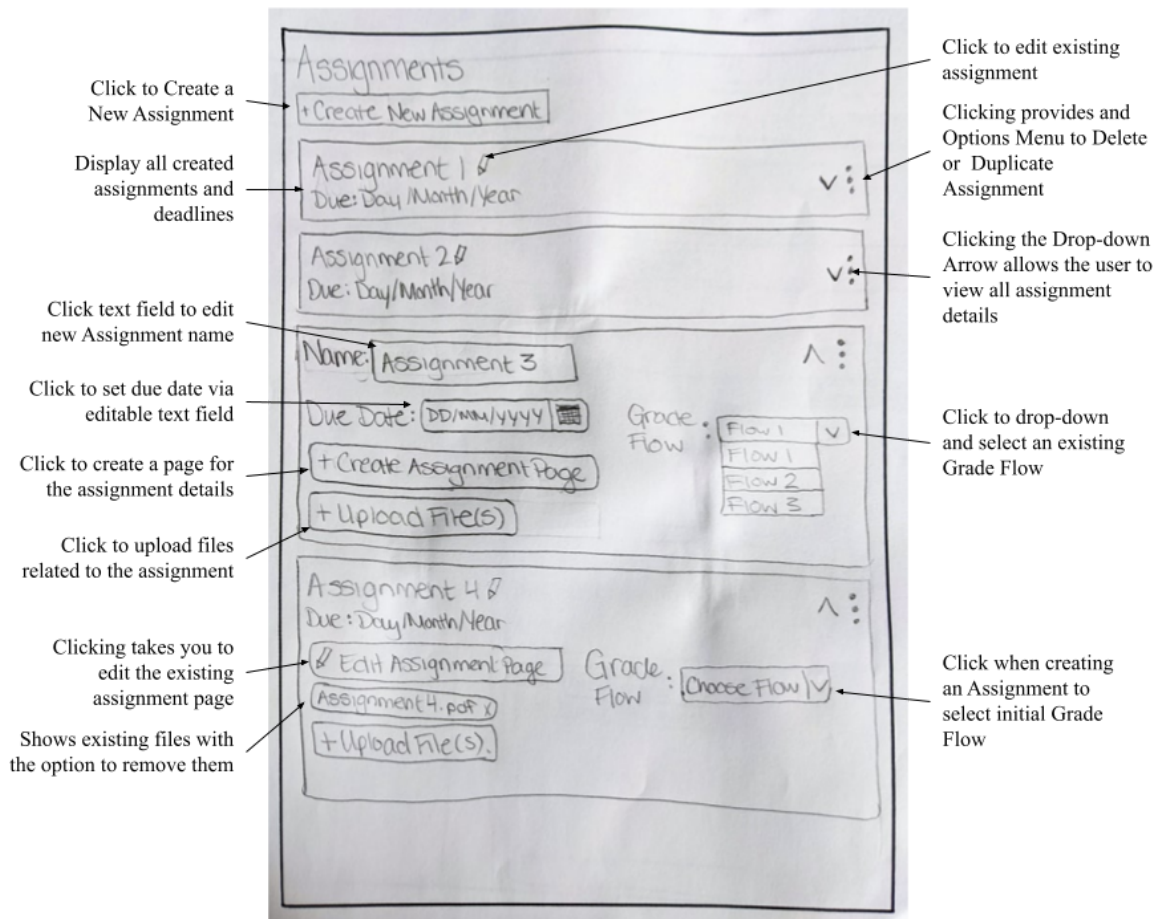


Figure 11: UI Sketch 6 - Assignments Page (Educator Side)

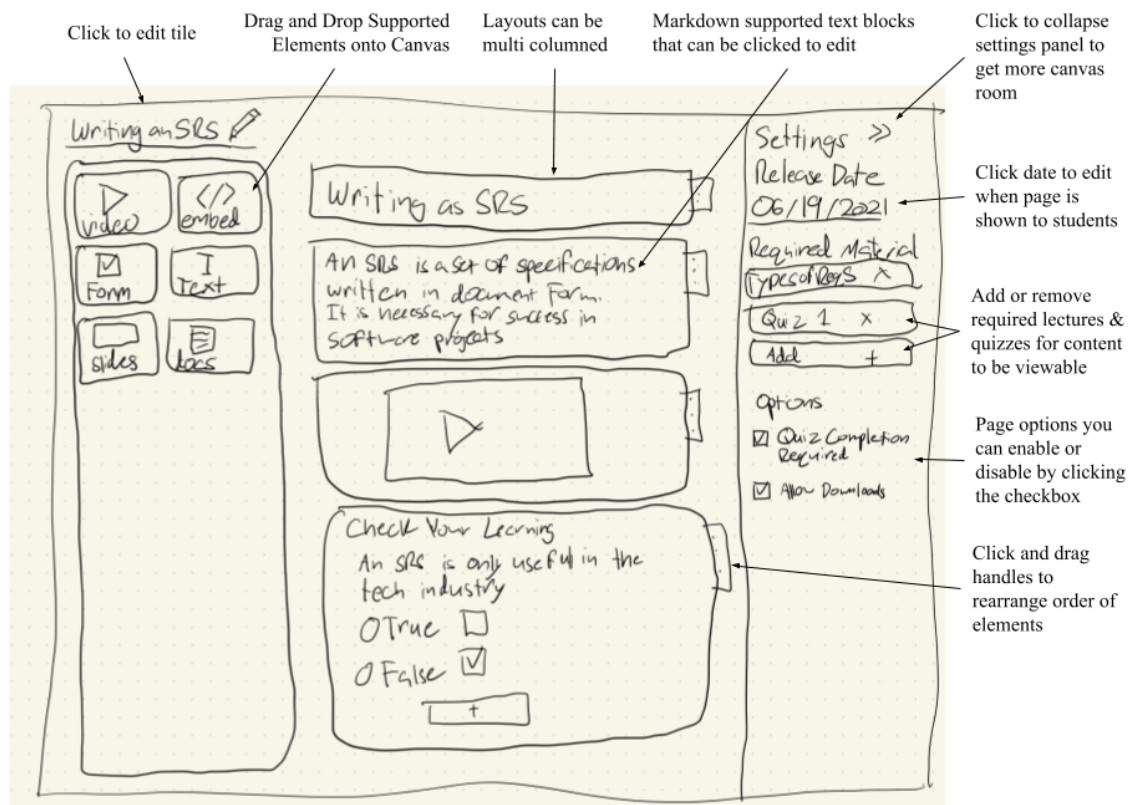


Figure 12: UI Sketch 7 - Edit Lecture Page

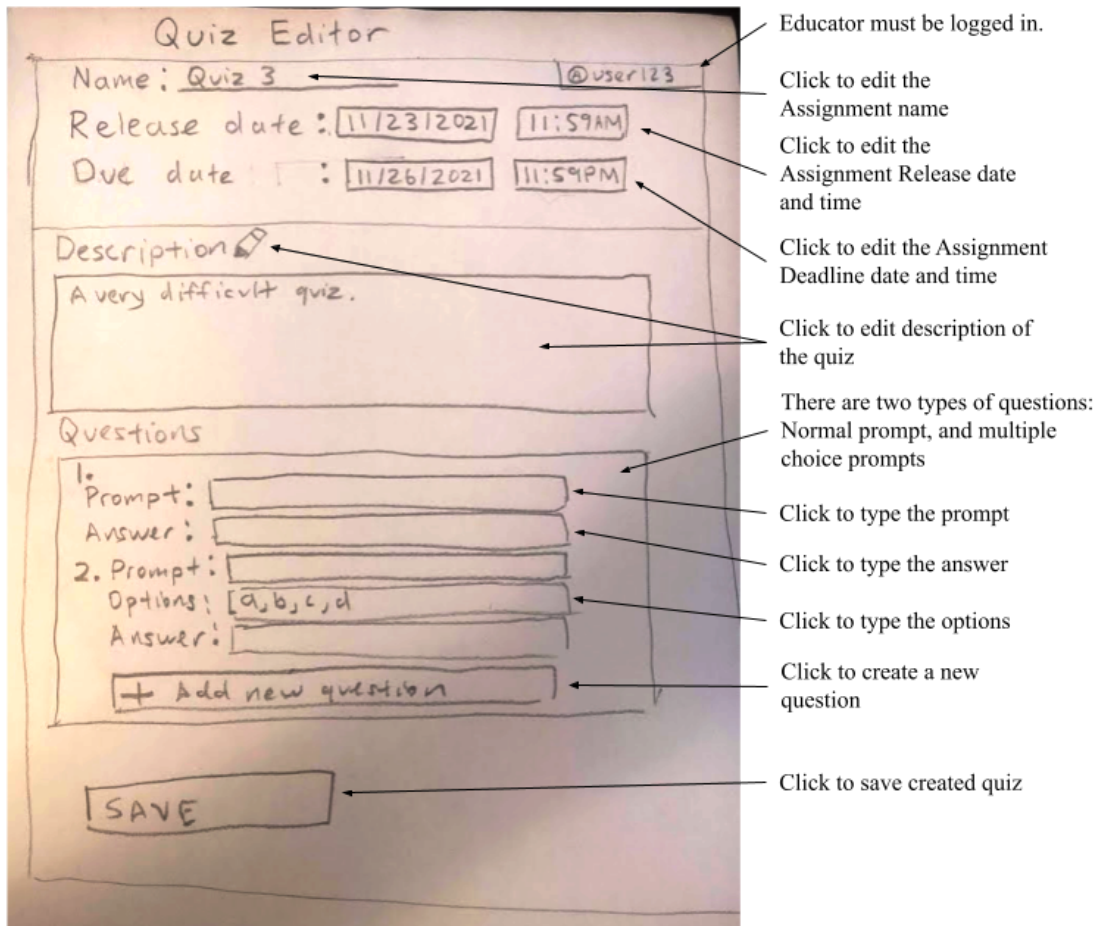


Figure 13: UI Sketch 8 - Edit Quiz Page

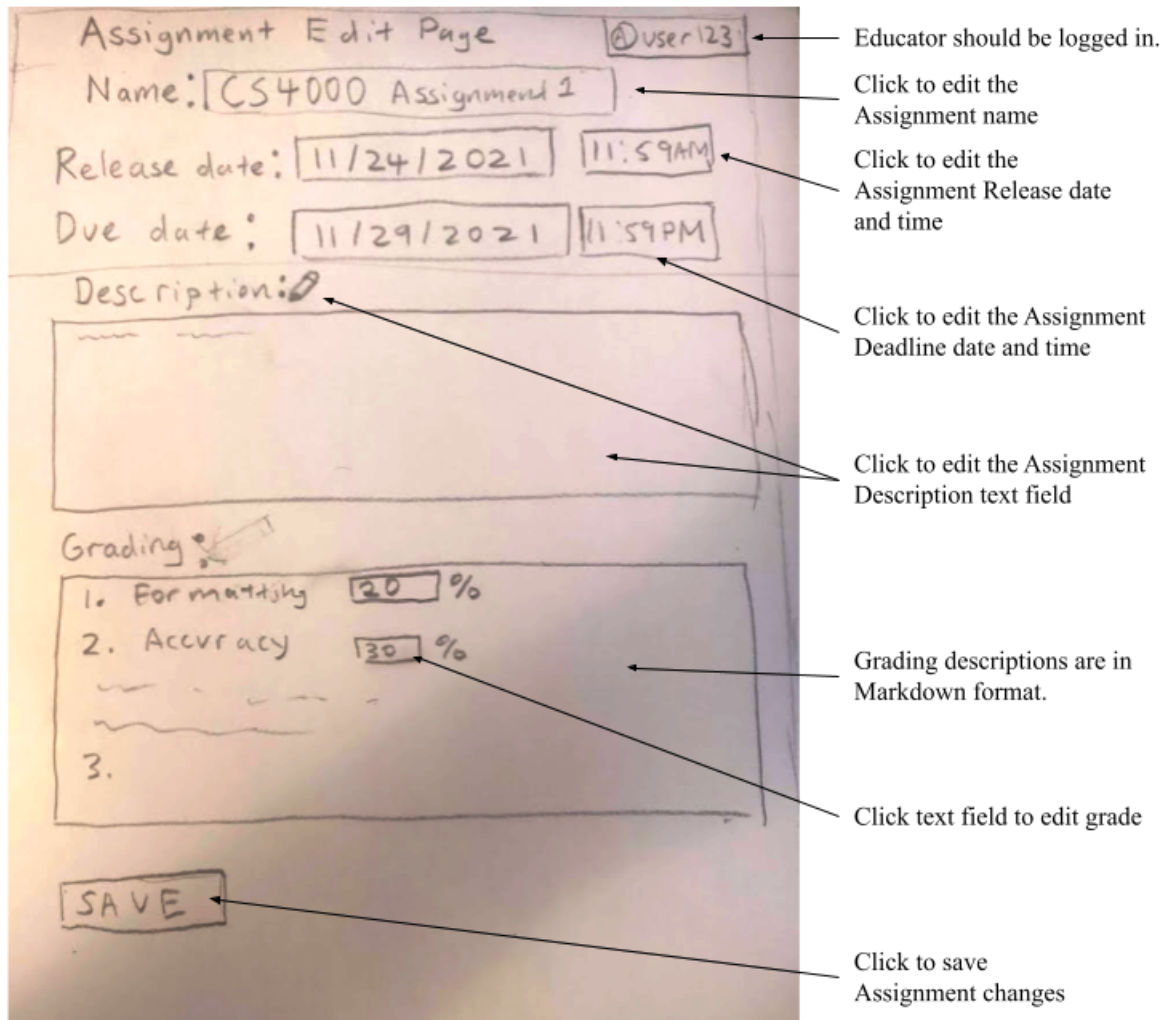


Figure 14: UI Sketch 9 - Edit Assignment Page

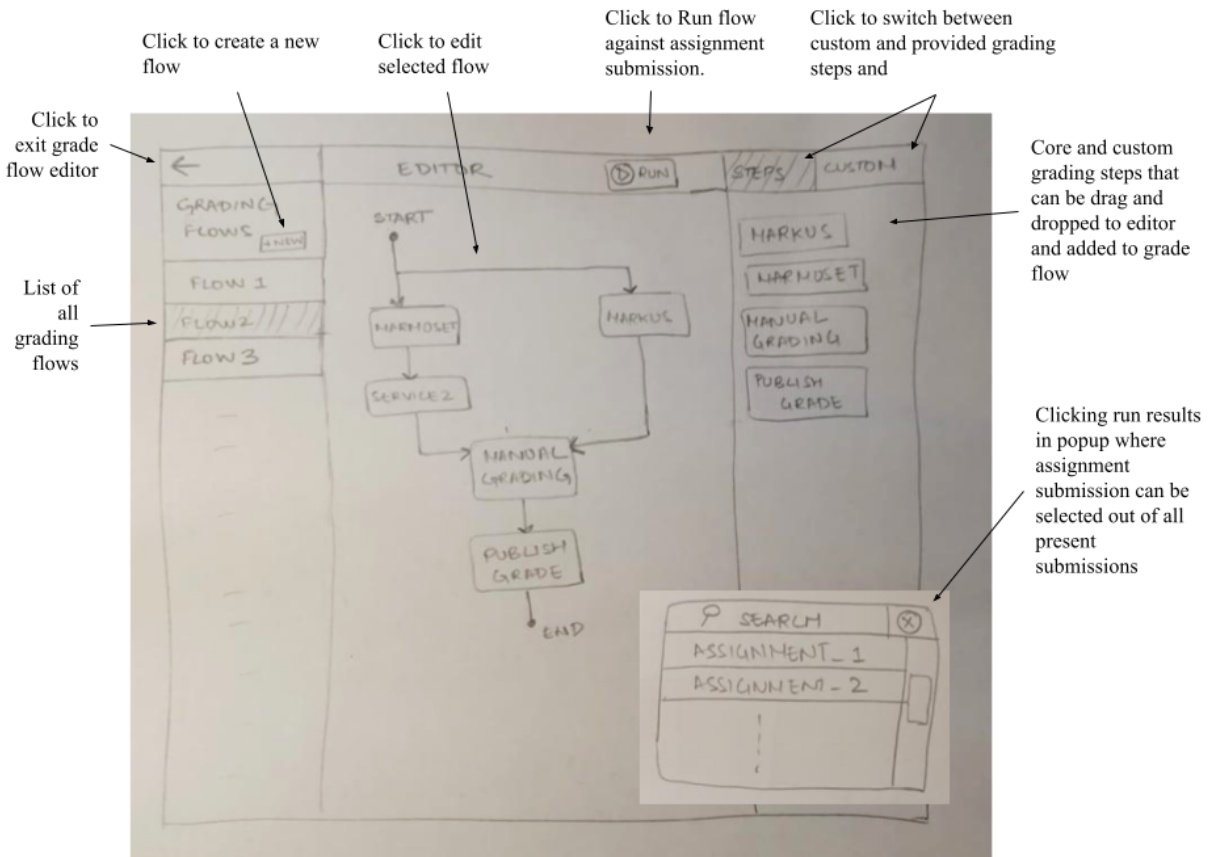


Figure 15: UI Sketch 10 - Edit Assignment Grade Flow

3.2 Scenarios and State Machines

This section contains the scenarios and state machine of the top three use cases.

Scenarios represent on full observable action's execution path through a use case, including the interactions between the system and any external actors.

For EduWizard, that is the interactions between the system and the user (e.g., CS Students and CS Educators), and the course site.

The state machine model represents the system's behaviour, and it can serve as a navigation diagram where transitions are triggered by input events and the UI screens are displayed as states in the model.

3.2.1 Scenarios

The following section includes scenarios and sub-scenarios pertaining to our top three use cases for EduWizard: Manage Course Site, Interact with Course Site, and Dispute Grade.

Scenario 1

Use Case Name: Manage Course Site

Trigger: CS Educator opens EduWizard platform

Preconditions:

- CS Educator has an EduWizard Account
- Site host has been procured

Stakeholders: CS Educator

Postconditions:

- CS Educator has updated the course site
- EduWizard has compiled and published the course site
- Course site is hosted on the site host

CS Educator	EduWizard	Site Host
1. Enters username and password and clicks the Sign in button		
	2. Authenticates user login using username and password	
3. Clicks "Creates New Course" button on the EduWizard Course landing page		
4. Initializes course site UI and layout		

	5. Compiles website with proper layout and sends it to site host	
		6. Hosts course site and makes visible to users
7. Verify that the course site has been successfully set up		
Alternative 1: Post lecture		
A1.1 Records a lecture		
A1.2 Enters username and password and clicks the Sign in button		
	A1.3 Authenticates user login using username and password	
A1.4 Uploads lecture video to EduWizard		
A1.5 Specifies where they want the lecture video to be displayed on course site		
	A1.6 Recompiles website to include the new uploaded lecture video and sends it to site host	
		Goto 6
Alternative 2: Post assignment		
A2.1 Creates a new assignment		
A2.2 Enters username and password and clicks the Sign in button		
	A2.3 Authenticates user login using username and password	
A2.4 Uploads assignment to EduWizard		
A2.5 Specifies the due date for the assignment		

	A2.6 Update assignment deadline	
A2.7 Specifies where they want the assignment to be displayed on course site		
	A2.8 Recompiles website to include the new assignment and updated assignment deadline and sends it to site host	
		Goto 6
Alternative 3: Update grades		
A3.1 Enters username and password and clicks the Sign in button		
	A3.2 Authenticates user login using username and password	
A3.3 Updates a grade in the EduWizard gradebook		
	A3.4 Recomputes class averages, individual averages and other grade metrics.	
	A3.5 Sends updated grades to the site host	
		Goto 6
Alternative 4: Upload quiz		
A4.1 Formulates a quiz		
A4.2 Enters username and password and clicks the Sign in button		
	A4.3 Authenticates user login using username and password	
A4.4 Creates quiz on EduWizard		

A4.5 Specifies quiz due date		
	A4.6 Updates quiz deadline	
A4.7 Specifies quiz time limit		
	A4.8 Updates quiz time limit	
A4.9 Specifies where they want the quiz to be displayed on course site		
	A4.10 Recompiles website to include the new quiz and updated deadline and sends it to site host	
		Goto 6
Exception 1: Incorrect login		
E1.1 Enters incorrect username or password and clicks the Sign in button		
	E1.2 Cannot authenticate login and informs user that their username/password is incorrect	
E1.3 Enters correct username and password and clicks the Sign in button		
	Goto 2	
Exception 2: Course site is down		
	E2.5 Compiles website and forwards to the site host	
		E2.6 No response
	E2.7 Informs user that site host is down, and course site is unable to be displayed	
	E2.8 While site host is down, continue polling site host and attempt to send compiled website to site host	

		E2.9 Acknowledge receipt of compiled website and hosts course site
	E2.10 Informs user that course site is back up and compiled website has been successfully sent to the site host	
Goto 7		

Scenario 2

Use Case Name: Interact with Course Site

Trigger: CS Student opens Course Site

Preconditions:

- CS Student has a course account

Stakeholders: CS Students

Postconditions:

- CS Student has completed an assignment or quiz, or viewed an announcement, lecture or grade

CS Student	Course Site	Site Host
1. Enters username and password and clicks the Sign in button		
	2. Authenticates user login using a username and password	
		3. Retrieve course home page
	4. Display home page	
5. Navigate to “Lectures” section of the course site		
	6. Fetch selected page from the site host	
		7. Retrieve “Lectures” page
	8. Display “Lectures” page	

9. Select lecture to view from the list available on the course site		
	10. Fetch selected lecture from the site host	
		11. Retrieve selected lecture video
	12. Display selected lecture	
13. View lecture		
14. Pause, play, fast forward, or rewind lecture		
	15. Receive action and modify lecture as appropriate	
15. Finish viewing lecture		
16. Sign-out and close course site		
Alternative 1: View course announcements		
A1.5 Navigate to the “Announcements” section of the course site		
	A1.6 Fetch selected page from the site host	
		A1.7 Send “Announcements” page
	A1.8 Display “Announcements” page	
A1.9 Select desired announcement to view		
	A1.10. Fetch selected announcement from the site host	
		A1.11 Send selected announcement

	A1.12 Display selected announcement	
A1.13 View announcement		
A1.14 Go-to 16		
Alternative 2: View Course Grades		
A2.5 Navigate to the “Grades” section of the course site		
	A2.6. Fetch selected page from the site host	
		A2.7 Send “Grades” page
	A2.8 Display “Grades” page	
A2.9 Select desired grade to view		
	A2.10 Fetch selected grade from the site host	
		A2.11 Send selected grade and grade feedback
	A2.12 Display selected grade and grade feedback	
A2.13 View grade and grade feedback		
Goto 16		
Alternative 3: Complete assignment		
A3.5 Navigate to the “Assignments” section of the course site		
	A3.6 Fetch selected page from the site host	
		A3.7 Send “Assignments” page

	A3.8 Display “Assignments” page	
A3.9 Select desired assignment to view		
	A3.10 Fetch selected assignment from the site host	
		A3.11 Send selected assignment details
	A3.12 Display selected assignment	
A3.13 View, complete, and prepare assignment for submission		
A3.14 Navigate to “Submissions” page		
	A3.15 Fetch selected page from the site host	
		A3.16 Send Submissions page
	A3.17 Display Submissions page	
A3.18 Select desired assignment to submit		
	A3.19 Fetch selected assignment from the site host	
		A3.20 Send selected submission box
	A3.21. Display selected submission box	
A3.22 Upload assignment to the submission box		
	A3.23 Send assignment to the site host	

		A3.24 Save submission
	A3.25 Display confirmation of submission page	
Goto 16		
Alternative 4: Complete quiz		
A4.5 Navigate to the “Quizzes” section of the course site		
	A4.6 Fetch selected page from the site host	
		A4.7 Retrieve “Quizzes” page
	A4.8 Display “Quizzes” page	
A4.9 Select desired quiz to complete		
	A4.10 Fetch selected quiz from the site host	
		A4.11 Retrieve selected quiz
	A4.12 Display selected quiz	
A4.13 Start quiz		
	A4.14 Send quiz start time to the site host	
		A4.15 Log quiz start time
	A4.16 Display quiz	
A4.17 Answer quiz questions and review responses		
A4.18 Submit completed quiz		
	A4.19 Send quiz submission to the site host	

		A4.20 Retrieve submitted quiz answers
	A4.21 Display quiz complete page	
Goto 16		
Exception 1: Assignment upload fails		
	E1.23 Assignment upload fails	
	E1.24 Display “Incomplete Submission” page	
E1.25 While assignment fails to upload continue attempting to upload		
	Goto A3.23	
Exception 2: Quiz time runs out before submission is made		
	E2.23 Quiz is automatically submitted	
		Goto A4.20

Scenario 3

Use Case Name: Dispute Grade

Trigger: CS Student receives a graded assignment, which they believe is marked incorrectly

Preconditions:

- CS Student submitted an assignment.
- CS Teaching Assistant receives the assignment submitted by the CS Student.
- CS Teaching Assistant completes marking of the assignment and updates the grade.

Stakeholders: CS Student, CS Teaching Assistant

Postconditions:

- The grade is updated if needed and the CS Teaching Assistant comments are visible
- CS Student is notified of the conclusion of the grade dispute

CS Student	Course Site	Teaching Assistant
1. Select assignment to dispute and create a re-mark request.		
2. Add evidence and comments to the request and click submit.		
	3. Receives submission from student account.	
	4. Posts the re-mark request the Educator facing side of the course site.	
	5. Notifies CS Teaching Assistant who graded the assignment.	
		6. Grader account receives access to the comments and evidence.
		7. Updates assignment grades and marks as complete.
	8. Receives updated remark request.	
	9. Posts the remark request to the course site with updated status and assignment grade.	
	10. Notifies the student account that created the re-mark request with updates to the request.	
11. Receives notification on their student account.		
Alternative 1: CS Teaching Assistant rejects the re-mark request		
		A1.7 Adds comments to the remark request explaining why they are rejecting the request and marks the request as "Rejected".

	A1.8 Receives updated remark request from grader account.	
	A1.9 Posts the remark request to the course site with updated comment and status.	
	Goto 10	
Alternative 2: CS Teaching Assistant rejects re-mark request and deducts grade		
		A2.7 Deducts grade from assignment attached to re-mark request where surplus grades were given.
		A2.8 Adds comment explaining why they rejected the remark request and the grade deduction.
		A2.9 Marks re-mark request as “Grade Amended”.
	Goto 8	
Alternative 3: CS Student cancels re-mark request before completion		
A3.7 Opens re-mark request on course website.		
A3.8 Clicks “Cancel remark request”		
	A3.9 Receives updated re-mark request.	
	A3.10 Posts re-mark request with cancelled status.	
	A3.11 Notifies the teaching assistant responsible for remark request of cancelled status.	
		A3.12 Receives the notification that the remark request has been cancelled.

	A3.13 Notifies the student account that cancellation was successful	
Goto 11		
Alternative 4: CS Teaching Assistant doesn't respond to the re-mark request notification for one week		
	A4.6 Tracks that no activity has occurred on re-mark request for a week.	
	A4.7 Sends a notification to all CS Teaching Assistant accounts in the course - "Re-mark request waiting for response"	
		A4.8 Assigns the re-mark request to themselves.
		Goto 6
Exception 1: CS Teaching Assistant closes EduWizard application while updating the request		
		E1.7 Starts editing a field in the re-mark request.
	E1.8 Auto-saves progress when field value is updated on the re-mark request user interface.	
		E1.9 Closes application.
		E1.10 Reopens application
	E1.11 Restores the re-mark with saved values.	
		E1.12 Finishes re-mark request evaluation
		Goto 7
Exception 2: Student device loses access to the internet while submitting re-mark request		
	E2.3 Save progress on remark request.	

	E2.4 Display error message notifying user request was not submitted and unable to access the internet	
E2.5 Restores internet connection to device		
E2.6 Clicks submit request		
	Goto 3	

3.2.2 State Machine

Below is the navigation diagram and descriptions for EduWizard of the top three use cases. The states in the navigation diagram correspond with the UI screens, and the transitions are triggered by input events (e.g., button clicks). This Navigation Diagram is divided into three key diagrams. The first diagram is the superstate that encompasses both the CS Educator and CS Student sides of our application. This contains the initial state where the user will login and the final state where the user logs out.

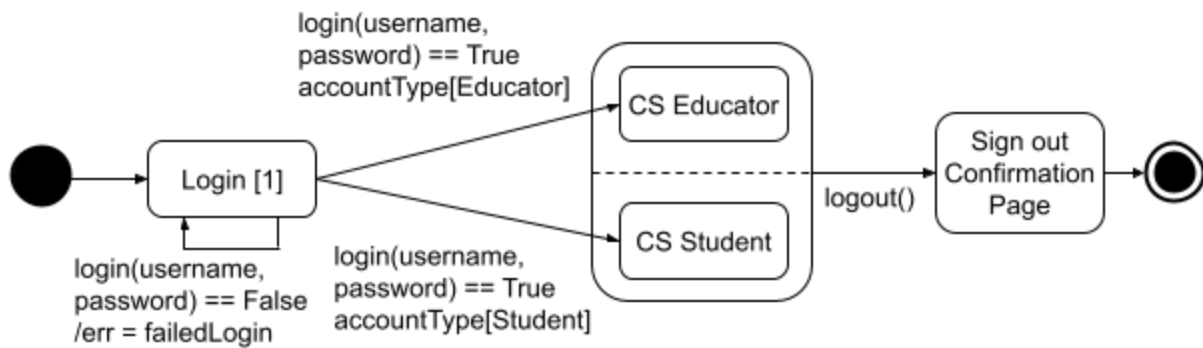


Figure 16: Navigation Diagram 1 - Superstate

For the sake of clarity we divided our two superstates into separate diagrams, though they run concurrently as seen in our main diagram.

The first superstate diagram is for the CS Educator superstate. This superstate is entered via a successful login from an Educator account type and exited via a logout action from any page.

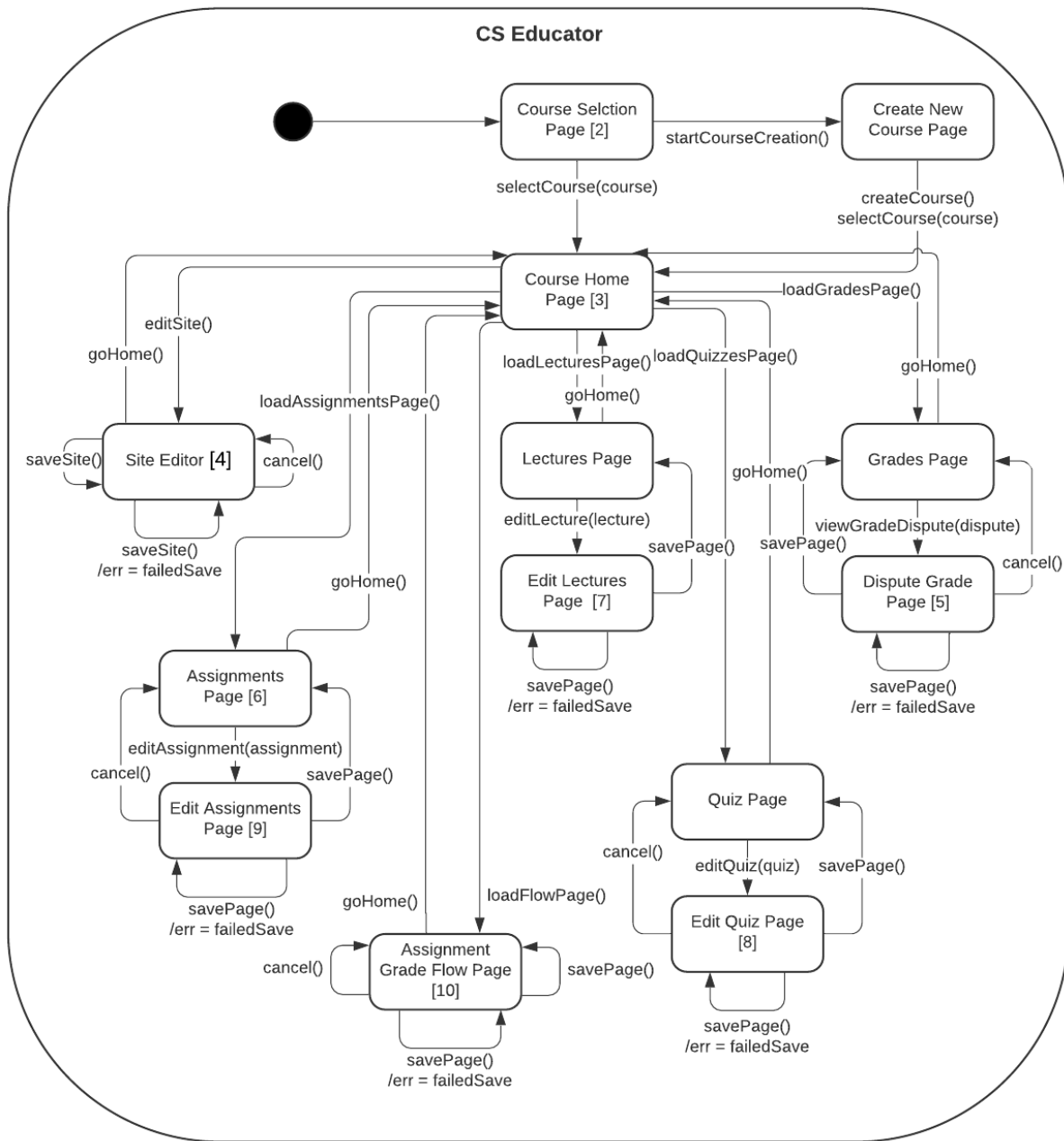


Figure 17: Navigation Diagram 2 - CS Educator Superstate

The second superstate diagram is for the CS Student superstate. This superstate is entered via a successful login from a Student account type and exited via a logout action from any page.

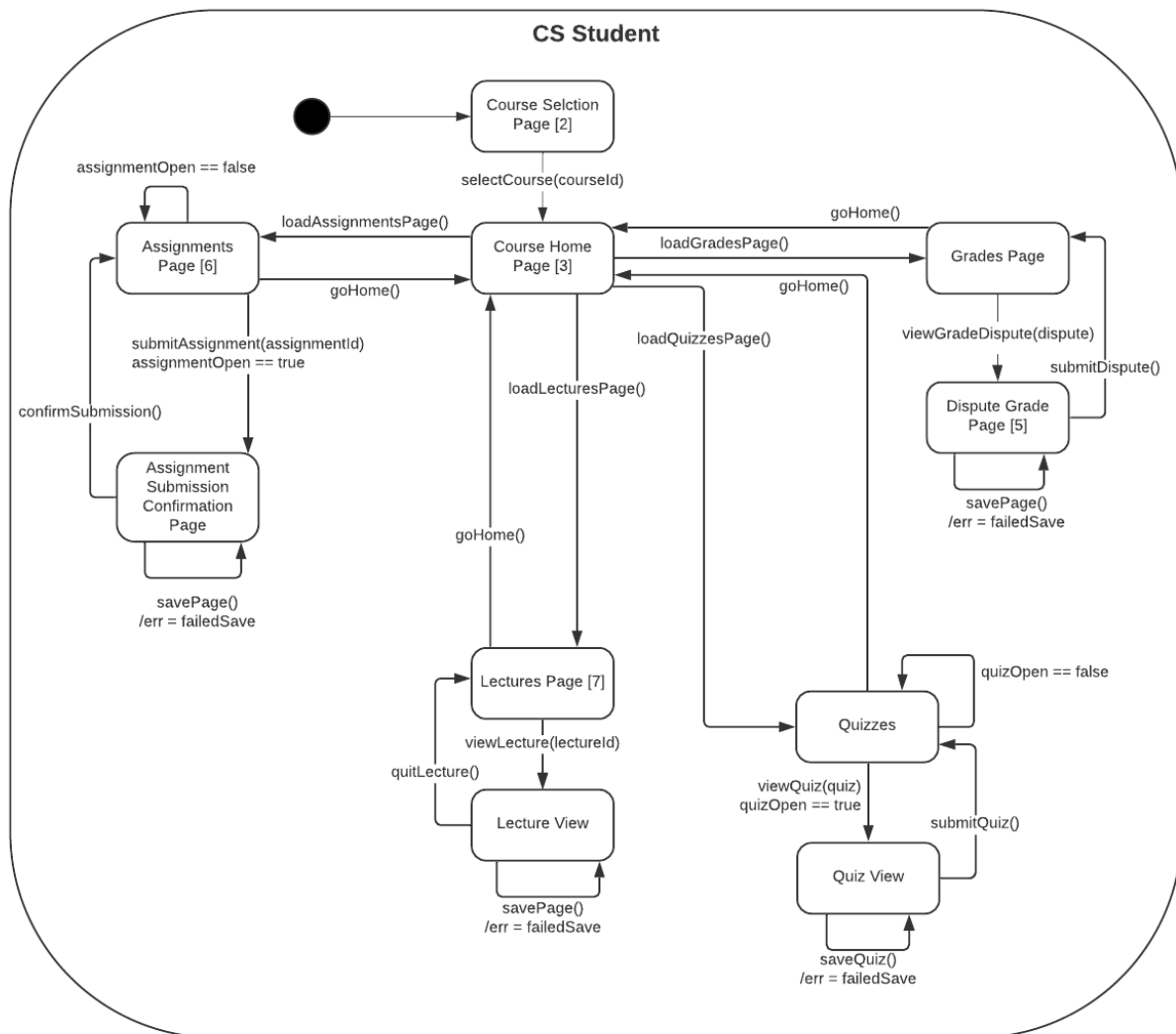


Figure 18: Navigation Diagram 3 - CS Student Superstate

Below are the descriptions of the variables and functions used within each superstate and the functionality of each state.

Variables Legend:

/err = failedLogin

Error message sent by the system if login(username, password) indicates it does not have a match for the provided username/password combination. It notifies the user their login was not successful.

/err = failedSave

Error message sent by the system if savePage() or saveSite() fails that notifies the user their changes are not saved.

accountType

An enum assigned to every account that indicates if it is a “Student” or “Educator” account. This determines which side of the course site the user has access to.

assignmentOpen

Variable that is set by the system based on the assignment deadline set by the CS Educator. If the assignment deadline has not passed and the assignment is open to submissions it is “true”, otherwise it is “false”.

quizOpen

Variable that is set by the system based on the quiz deadline set by the CS Educator. If the quiz deadline has not passed and the quiz is open it is “true”, otherwise it is “false”.

Transitions Legend:

cancel()

User clicks the “Cancel” button on any editable page (ex. Edit Assignments Page) and it discards their changes and navigates the user out of the editing screen.

confirmSubmission()

User clicks the “Confirm/Submit” button inside the “Assignment Submission” page.

createCourse()

Creates a blank site for the user to begin editing.

editAssignment(assignment)

User clicks on an assignment to edit and is navigated to the editable page for that assignment. (Screens 6, 9)

editLecture(lecture)

User clicks on a lecture to edit and is navigated to the editable page for that lecture. (Screens 6, 9)

editQuiz(quiz)

User clicks on a quiz to edit and is navigated to the editable page for that quiz. (Screens 6, 9)

editSite()

User clicks on the “Edit Site” icon, from the UI and navigates the editable page for the site. (Screens 3, 4)

goHome()

User clicks the “Home” icon, from their page to return to the Home Screen. (Screen 3)

loadAssignmentsPage()

User clicks the “Assignments” link on the course home page. (Screen 6)

loadFlowPage()

User clicks on the “Grade Flow” link and the Grade Flow page is loaded for the user to see. (Screen 3)

loadGradesPage()

User clicks on the “Grades” tab and the Grades page is loaded for the user to see. (Screen 3)

loadLecturesPage()

User clicks on the “Lectures” tab and the Lectures page is loaded for the user to see. (Screens 3, 7)

loadQuizzesPage()

User clicks on the “Quizzes” link and the Quizzes page is loaded for the user to see. (Screen 3)

login(username, password)

User clicks login which authenticates the user, with the username and password provided, the function returns “true” if the sign-in was successful (the username/password combination was valid) and “false” otherwise. (Screen 1)

logout()

User clicks the “Sign Out” link provided on the Home Screen to sign out of the course (Screen 3), or closes the course site from any page in the editor.

quitLecture()

User clicks the “Quit” button provided in the lecture.

savePage()

User clicks on the “Save” button to save the page changes. (Screens 8, 9, 10)

saveSite()

User clicks on the “Save” button to save the site layout changes. (Screens 8, 9, 10)

selectCourse(course)

User clicks the corresponding course button they are intending to interact with. (Screen 2)

startCourseCreation()

User clicks the “+” icon button, with the intention to create a new course. (Screen 2)

submitAssignment(assignmentId)

User uploads an assignment file for an assignment and clicks the “Submit” button.

submitDispute()

User clicks the “Confirm/Submit” button inside the “Grade Dispute” page to submit their grade dispute.

submitQuiz(quiz)

User completes a quiz and clicks the “Submit” button.

viewGradesDispute(dispute)

User clicks on the “Grade Dispute” link inside of the “Grades” page. (Screen 5)

viewLecture(lectureId)

User clicks the lecture link, from the “Lectures” page.

viewQuiz(quizId)

User clicks the quiz link from the “Quizzes” page.

States Legend:**Main Superstate:**

The main superstate is a concurrent state with one half dedicated to the CS Educator flow and the other to the CS Student flow. These superstates are entered via transition from the Login state, depending on the account type used at login.

CS Educator Superstate

The superstate that is entered when the user logs in with an Educator account type.

CS Student Superstate

The superstate that is entered when the user logs in with a Student account type.

Login

The state that is entered from the initial state that prompts the user to sign in. It collects account type information to determine if the CS Educator or CS Student superstate should be entered.

Sign out Confirmation Page

The state that is entered when the user clicks the “Sign Out” link provided on the Home Screen to sign out of the course (Screen 3). It displays a screen that confirms the user is signed out..

CS Educator SuperState:

The superstate that specifies the behaviour of our application when the user signs in with an Educator account type.

Assignment Grade Flow Page

The state where CS Educators are when they are viewing or editing assignment grade flows.

Assignments Page

The state where CS Educators can view all their assignments.

Course Home Page

The state where CS Educators can view all their course content, and navigate into specific content types.

Course Selection Page

The state where CS Educators can choose a course to view or edit.

Create New Course Page

The state entered when a CS Educator chooses to create a new course site, not a visible page.

Dispute Grade Page

The state where CS Educators can view, settle, and comment on Grade Disputes.

Edit Assignments Page

The state where CS Educators can edit their assignments' content.

Edit Lectures Page

The state where CS Educators can edit their lectures' content.

Edit Quiz Page

The state where CS Educators can edit their quizzes' content.

Grades Page

The state where CS Educators can view CS Student grades.

Lectures Page

The state where CS Educators can view and select from their existing lectures.

Quiz Page

The state where CS Educators can view and select from their existing lectures.

Site Editor

The state where CS Educators can edit their site layout.

CS Student Superstate:

The superstate that specifies the behaviour of our application when the user signs in with a Student account type.

Assignments Page

The state that shows the list of assignments that CS Students can select.

Assignment Submission Confirmation Page

The state that allows you to submit and get confirmation of whether the submission has been properly sent.

Course Home Page

The state that allows CS Students to navigate quizzes, assignments, grades and lectures.

Course Selection Page

The state that allows CS Students to pick what course they are trying to interact with.

Dispute Grade Page

The state that allows CS Students to dispute a grade for an assignment or quiz grade they have received.

Grades Page

The state that allows CS Students to view their grades for all course content.

Lectures Page

The state that allows CS Students to view all lectures that are available to them for that course, via multiple links

Lecture View

The state that allows CS Students to view a lecture and its files associated with it.

Quizzes

The state that allows CS Students to view the list of all course quizzes

Quiz View

The state that allows CS Students to view/do a quiz on the platform.

3.3 Quality Requirements

The top three quality requirements for our application are Usability, Performance and Reliability. We performed quality requirements interviews by applying the 100-dollar prioritization chart to obtain our key quality requirements.

The following chart is the result of the 100-dollar prioritization interviews:

	Quality Requirements					
Scores	Accuracy/ Precision	Usability	Performance	Reliability	Portability	Security
CS Student 1	20	20	30	10	5	15
CS Student 2	20	10	15	25	15	15
CS Educator 1	30	20	10	10	10	20
CS Educator 2	0	30	15	25	10	20
CS Educator 3	20	20	20	20	5	15
Sum	90	100	90	90	45	85

Following is a description of each requirement and how we plan on measuring them:

Usability: The site builder needs to be intuitive to use. The site builder should have easy-to-understand tools that are based upon similar tools already encountered by our user base. Educators should be able to easily update the site and manage their course content. The user interface should be responsive to user input and result in higher user engagement.

Measurement: Percentage of support requests made to the EduWizard support desk by educators with questions about platform usage (0% = no support requests were related to usability, 100% = all support requests were related to usability)

Performance: Performance for our application is important because online schools must support as many people as possible. This means people with slower internet connections should be able to load the page in a reasonable amount of time. We can accomplish this by creating pages that have a low overall bundle size.

Measurement: Bundle size of software in Megabytes, where lower is better

Reliability: The site builder also should be available, almost always. Since instructors may add material to their courses quite often over the span of a term, having downtimes can hinder the online learning experience. Low downtimes will assist in reliability.

Measurement: Downtime in minutes per year

We had a tie for second place, three requirements had a 90, so we used the verbal responses given by interviewees and the range of values to determine which two would be the actual second and third requirements. Given the verbal responses, and the fact that Accuracy/Precision received the only zero in our table it was decided that Usability, Performance, and Reliability would be the top three requirements.

Following is the rich fit criteria table elicited from the quality requirement interview:

	Usability	Reliability	Reliability
Minimum (Average and Standard Deviation)	29% $\pm$ 0.19%	3072 $\pm$ 2172.22 MB	5146.67 $\pm$ 4268.08 mins
Target (Average and Standard Deviation)	16% $\pm$ 0.12%	2048 $\pm$ 1448.15 MB	1753.33 $\pm$ 2641.69 mins
Outstanding (Average and Standard Deviation)	8% $\pm$ 0.08%	1024 $\pm$ 724.08 MB	286.67 $\pm$ 445.57 mins

The rich fit criteria table summarizes the outstanding, target, and minimum criteria of our top three quality requirements. It contains the mean and standard deviation, and the top three use cases determined by tallying up all the results from the 100 dollar prioritization exercise.

For the rich fit criteria, our team performed several quality requirement interviews by asking the interviewees their outstanding, target, and minimum criteria for each of the three requirements, then we calculate the mean and standard deviation to achieve the final result.

References

G. Thomson, “Accessibility for Ontarians with disabilities act (AODA),” Accessibility for Ontarians with Disabilities Act (AODA), 22-Apr-2014. [Online]. Available: <https://www.aoda.ca/>. [Accessed: 10-Dec-2021]

W3C, Web content accessibility guidelines (WCAG) 2.0, 11-Dec-2008. [Online]. Available: <https://www.w3.org/TR/WCAG20/>. [Accessed: 10-Dec-2021]